Graph Representation Learning

Marcelina Marjankowska

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Please use the official LaTeX template to type your answers available in Moodle. Please respect the notation from Table 1 in your answers whenever applicable:

Table 1: Notation.

σ	An element-wise non-linearity.
t	Iteration, or layer t.
$d^{(t)}$	The dimension of a vector at iteration t.
d	The dimension of a vector, and an abbreviation for $d^{(0)}$.
$\mathbf{1}^d \in \mathbb{R}^d$	A d-dimensional vector of all 1's.
$\mathbb{I}^d \subseteq \mathbb{R}^d$	The set of d-dimensional one-hot vectors.
$\mathbb{B}$	Boolean domain $\{0, 1\}$.
$\mathbf{b}^{(t)} \in \mathbb{R}^d$	A bias vector.
$\mathbf{x}_u \in \mathbb{R}^d$	The feature of a node $u \in V$.
$\mathbf{h}_u^{(t)} \in \mathbb{R}^{d^{(t)}}$	The representation of a node $u \in V$ at layer t.
$\mathbf{z}_u = \mathbf{h}_u^{(T)} \in \mathbb{R}^{d^{(T)}}$	The final representation of a node $u \in V$ after T layers/iterations.
$\mathbf{W}_x^{(t)} \in \mathbb{R}^{d^{(t+1)} \times d^{(t)}}$	Learnable parameter matrix at layer t.
MLP	A multilayer perceptron with ReLU as nonlinearity.

Question 3

The research question: The question I want to explore is whether we empirically observe an analogous to zero-one law on Barabási–Albert (BA) random graphs.

Motivation: The GCN-RNI (Graph Convolutional Networks with Random Node Initialization) demonstrated universal expressiveness on bounded-size graphs [1]. Conversely, the Zero-One Law paper [2] imposed an upper limit on their expressiveness: the architecture can solely approximate functions adhering to the zero-one law on Erdős–Rényi (ER) graphs. It also highlighted the model's capability to capture any graph invariant adhering to the given law on graphs with p=0.5. Generalizing this result to arbitrary p would provide a comprehensive understanding of these models' expressive power. In general, as argued in lectures, the graph cases in which MPNNs fail in expressiveness, are very regular, making these limitations quite irrelevant in practice. Here, we want to explore if empirically we observe a similar zero-one law pattern for a more realistic random graph model. If the asymptotic behaviour persists, the expressiveness constraint of MPNN-RNIs could potentially show in practice.

Overall approach: We will consider the Barabási–Albert (BA) random graph model [4] which is used for modelling real-life data including social networks, the World Wide Web and citation networks. It is based on the preferential attachment mechanism where nodes with higher degrees are more likely to continue receiving more connections. The BA algorithm is mostly used in modelling growing networks and has several variants [5, 3].
For the model, I will consider GCN-RNI and sum aggregation because of its expressive power [1, 10, 7]. I will then draw models with random features and evaluate them on random graphs of increasing size. We will then observe what proportion of the graphs was evaluated by the model to one. We are interested in seeing an asymptotic behaviour of this number to one or zero.
The following experiments are designed to address the following questions: Do we empirically observe a zero-one law on BA graphs? Could it have implications on predictions done for real-life datasets? Do we empirically observe a zero-one law for node-level prediction on BA graphs?
I will start with duplicating some of the results from the [2] paper. Then, I will evaluate more realistic network models. I will also evaluate models on the Cora ML citation network dataset and observe if the models output agrees with its asymptotic behaviour. Finally, I will also examine the model's performance on the node binry classification task. The code made is available at `https://github.com/sdfghjklsdfghjkldfghjkl/grl_miniproject`.

Experimental Setup: The experiments need to be conducted carefully so that requirements from

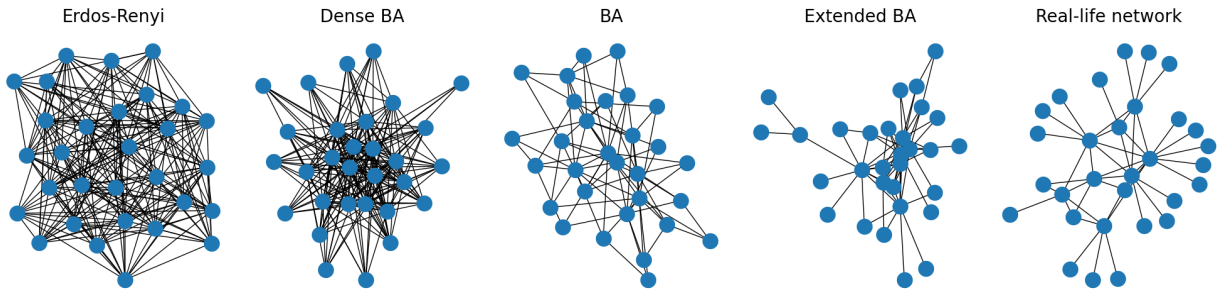


Figure 1: Graphs on 30 vertices generated using different random models. The last plot is an example of a real-life network, an internet graph.

[2] Theorem 4.6 are satisfied. For that, I will closely follow the implementation of the paper. For the evaluation data, I will use models from the networx library. We will consider **Erdos-Renyi** with $p = 0.5$, **Dense BA** with parameter $m = n/2$, **BA** with $m = 3$ and **Extended BA** with $m = 1, p = 0.5, q = 0.1$ that incorporates rewiring edges. You can see example generations in Figure 1. For the real-life network I will use the $2,995$-node **Cora ML Dataset** from pytorch-geometric CitationFull.

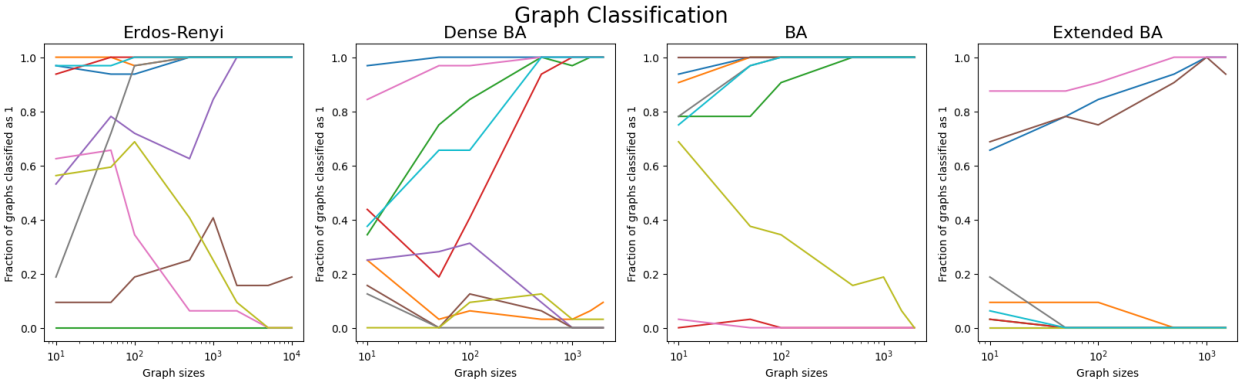


Figure 2: Each plot shows evaluations of 10 models on the different graphs. The y axis shows what fraction of the 32 generated graphs was evaluated to 1 by given model.

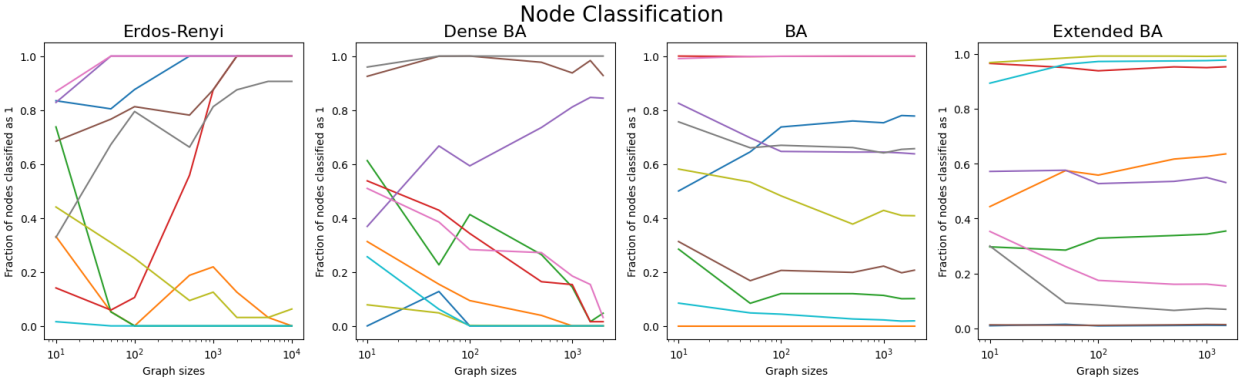


Figure 3: Each plot shows evaluations of 10 models on the different graphs. The y axis shows what fraction of all nodes from the 32 generated graphs were evaluated to 1 by the given model.

In the first experiment, we will use the models as binary classifiers. We fix 10 MPNN-RNI with sum aggregation, parameters drawn randomly from $U(-1, 1)$ and truncated sigmoid as non-linearity. I will set the number of layers to 3 with all layers of width 64. Mean-pooling will be applied to get the graph representation. It will be followed by a 2-layer MLP with a hidden layer size of 100 and sigmoid after last layer. For each graph model and graph size, we draw 32 instances. All of them are evaluated on the model and the graphs will be classified as 0 if the value is smaller than 0.5, and as 1 otherwise. For each graph size we will note the fraction of the 32 graphs that were evaluated to 1. In the second experiment for each of the graph models, we will draw 100 random MPNNs with parameters as before, and record whether their asymptotic behaviour was zero or one with the breaking point being at 0.5. Then, I will evaluate the Cora ML network on the models. In the third experiment, the setup is identical as before, with the exception of output layer of MPNN being 7 and the model being used for node classification.

Findings and Discussion: The experiment results can be seen in Figures 2 and 3. Plots correspond to different graph models and lines correspond to the random MPNN models. In the graph classification case, the y-axis values are ratios of graphs that were evaluated to 1 by the model. We can see in the first two plots the same behaviour as in the Zero-One paper with the ER graph converging slowly (compare with SumGNN, $L = 3$ plot from the paper). In the appendix of [2], the authors also consider the (Dense) BA model and observe the zero-one behaviour. However, they use an unrealistic parameter that produces a very dense graph with a small degree range and structural regularities. See Dense BA in Figure 1 for example. The next two plots in Figure 2 show that both the BA model with $m = 3$ and the Extended BA model also follows the zero-one asymptotic behaviour and the convergence rate is quick.
Looking at the evaluations of the Cora Dataset, we obtained that both the ER graph and the dense BA asymptotic behaviours in 30 out of 100 cases were different from the model prediction. This is a significant number that suggests that the asymptotic behaviour is correlated to the specific type of random graph model and not all graphs in general. Numbers 7 and 3 in the cases of the BA model and the Extended BA model are very interesting. Firstly, it suggests that in fact, the graphs produced by these models are more similar in structure to the citation network. Secondly, it means that the zero-one behaviour that we observe in these models could have practical implications.
Lastly, in figure 3 we can see models acting as node classifiers. The ER graphs have an asymptotic behaviour while BA and Extended BA seem constant. The result for the ER graph agrees with the theoretical result from [2] Lemma 4.11: there is some vector z such that the probability of SumGNNs last layer representation being z for all nodes in an n-vertex ER graph instance, tends to 1 as n tends to infinity. Meaning that the model will at some point start classifying all nodes to the same value. It's interesting to see that this no longer holds for the BA model, which is probably a consequence of nodes having more diverse neighbourhoods.

Conclusion, Limitations and Outlook: The experimental results reflect the theoretical bound on the expressive power introduced in [2]. We have also seen the zero-one behaviour in the case of more realistic network models. The evaluation of the Cora dataset suggests that the limitations could be reflected in real-life examples. However, experiments with node classification show that the inherited mechanisms in the case of ER and BA graphs are not the same. On top of that, MPNNs have been extremely successful in practice, see for example [9, 8, 6], so our speculations are dubious. To further clarify the matter one could try to follow a similar theoretical analysis as in [2] for the BA model. Also, the zero-one behaviour on ER graphs persists on trained networks, and empirical evaluation of that for both ER and BA graphs could be interesting.

References

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